

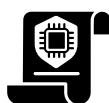
A MONTH AFTER Prime Minister Narendra Modi's state visit to the US in June 2023—when semiconductor major Micron announced plans to invest in India—he inaugurated the second edition of the annual Semicon India summit in Gandhinagar, Gujarat.

Since it came close on the heels of Micron's announcement of its intention to set up an ATMP (assembly, testing, marking, and packaging) unit, there were hopes that more would follow. There was just one; AMD's commitment to invest \$400 million to set up its largest design centre in Bengaluru. But India has yet to receive an application from a leading chip player for setting up a semiconductor fabrication unit. Media reports suggest that Tower Semiconductor has filed a fresh application, but the government is yet to confirm this.

Of course, India's dream of becoming a semiconductor hub isn't new. There have been failures aplenty over the past 60 years. But hopes soared that that script would change when, in December 2021, the government announced a production-linked incentive scheme worth ₹76,000 crore for chip and display fabrication units.

The timing seemed ideal since the announcement came in the middle of an acute shortage of chips globally. Timing apart, the government is aware of the challenges in this path and is taking the long view. Rajeev Chandrasekhar, Minister of State, Ministry of Electronics and IT (MeitY), says, "Semiconductor manufacturing is not a today or tomorrow game. It is a day after tomorrow and down the road game... decades-long game."

THE INDIAN SEMICONDUCTOR MISSION



STATES LIKE GUJARAT, UTTAR PRADESH AND ODISHA HAVE ANNOUNCED SEMICONDUCTOR POLICIES



\$2.75 BILLION-WORTH MICRON ATMP FACILITY ANNOUNCED AND APPROVED



APPLIED MATERIALS TO BUILD A COLLABORATIVE ENGINEERING CENTRE IN BENGALURU WITH AN INVESTMENT OF \$400 MILLION



AMD TO INVEST AROUND \$400 MILLION OVER THE NEXT FIVE YEARS TO CREATE ITS LARGEST DESIGN CENTRE IN BENGALURU



LAM RESEARCH TO TRAIN 60,000 INDIAN ENGINEERS IN NANOTECHNOLOGIES OVER 10 YEARS



BTECH IN VLSI DESIGN BY AICTE: MORE THAN 300 INSTITUTES TO OFFER UNDERGRADUATE AND POSTGRADUATE COURSES



FUTURE DESIGN: SEVEN FABLESS START-UPS APPROVED

The trouble is that other countries too saw this as a chance to bring chip production onto their shores. From the US to Europe, Japan, and even developing nations like Malaysia, a host of countries are attracting investments.

This poses a big challenge for India. That's because setting up a foundry is an expensive affair, costing anywhere between \$2 billion and \$20 billion, per industry estimates, and it takes six to eight years to break even. Besides, there are a whole host of factors that companies look at before zeroing in on a location. Helen Chiang, Country Manager for IDC Taiwan and IDC's Lead for Asia Semiconductor Research, highlights four key focus areas: infrastructure (including reliable power, water resources, transportation networks, and telecommunications); a skilled talent pool (in engineering, materials science, and electronics); proximity to major clients, supply chains, and target markets; lead times and shipping risks; and lastly, geopolitical stability. India's nascent chip ecosystem has some distance to go to meet these demands.

BAGGAGE OF THE PAST

The government may have sought a fresh start in December 2021. But it had to first deal with the considerable baggage of the past. India has not just lost out to developed nations but even to developing ones. For instance, Intel has picked Malaysia to build its largest 3D chip packaging facility.

It wasn't always like this. Way back in the 1960s and '70s, Intel had shown interest in starting a manufacturing unit in India, but that interest was not reciprocated. "Robert Noyce, the Co-founder of Intel, had visited India in 1969 to explore the opportunity to set up a semi-